

1 Source-only model selection and limits of fixed spectral-sharing pipelines
2 in cross-subject motor-imagery EEG decoding

3 Short title: Source-only selection and fixed spectral-sharing pipelines in motor-imagery EEG

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10

11 **Abstract**

12 We audited source-only training-duration selection and evaluated fixed shared mu/beta input
13 pipelines for cross-subject motor-imagery EEG decoding. An exploratory four-class study used
14 nine BNCI2014_001 participants, excluding both target sessions from each corresponding model
15 fit and duration selection. Separate frozen binary pipelines were evaluated on PhysioNet (109
16 people), Cho2017 (52) and Lee2019 offline-training runs (54, two sessions), without target fitting
17 or adaptation. We re-audited saved predictions and source partitions. A BNCI signal analysis,
18 specified after decoder outcomes were known, described baseline-relative mu/beta power and
19 associations with saved binary LOSO balanced accuracy. Mean-rank duration selection increased
20 internal balanced accuracy from 33.72% to 42.67% compared with mean-loss selection; S3 and S8
21 contributed 87.57% of the aggregate gain. A matched-runtime fixed-duration control improved
22 four of nine people despite a positive mean. Shared-minus-broad external differences were +0.573
23 percentage points (pp) on PhysioNet (95% person-bootstrap interval $[-0.097, 1.242]$), -1.513 pp on
24 Cho2017 $[-2.249, -0.804]$ and $+0.204$ pp on Lee2019 $[-0.515, 0.969]$. Cho2017 and Lee2019
25 Holm-adjusted permutation p-values were 0.000100 and 0.604. These complete-pipeline contrasts
26 used unequal source-selected durations. The BNCI signal analysis retained all 5,184 trials; its
27 2,592 hand-imagery trials gave mean signed laterality of -0.250 dB in mu and -0.168 dB in beta,
28 with signs varying between people. Training-duration choices yielded large but concentrated gains
29 in this development benchmark. Shared-input pipelines showed no consistent external advantage.
30 BNCI sensor patterns provide descriptive context; nine-person associations do not establish a
31 decoder mechanism. Unknown voltage calibration and offline evaluation limit deployment
32 conclusions.

33 **Introduction**

34 Motor-imagery EEG decoding must distinguish imagined actions in scalp recordings whose spatial
35 and spectral structure varies between people. A calibration-free model should transfer to a new

36 person without fitting to that person’s labels or distributional statistics. This requirement differs
37 from within-person evaluation and adaptation using unlabeled target recordings. BCI Competition
38 IV dataset 2a is a small, well-documented benchmark [1,2], but its nine participants make the
39 cohort mean especially sensitive to a few large changes. Reviews and multi-dataset evaluation
40 frameworks document variation between datasets and show how evaluation design shapes reported
41 performance [3,4].

42 A decoder that predicts predominantly one class can have balanced accuracy near chance even
43 when the same architecture performs well after a different training duration. This failure may be
44 attributed to an input distribution that requires normalization or a different representation,
45 overlooking the role of model selection. Likewise, an increase in average balanced accuracy may
46 reflect recovery in one or two participants rather than consistent improvement across people.

47 Source-only fitting prevents target leakage within each fit, but it does not provide independent
48 confirmation when the research programme was developed after examining the same benchmark.

49 We address three questions. First, can heterogeneous source-validation loss curves lead to very
50 short training schedules, and does within-fold rank aggregation change those schedules? Second,
51 how does a pipeline that processes predefined mu/beta views with one shared EEGNet compare
52 with broad input, alongside spatial, capacity, and robustness controls? Third, how do the complete
53 source-selected pipelines perform in external binary transfer? A separate descriptive analysis
54 characterizes measured BNCI sensor signals without changing the decoders. The external evidence
55 comprises an earlier PhysioNet evaluation [5,6] and a subsequently frozen, separately fitted
56 pipeline for Cho2017 [7,8] and Lee2019 [9,10]. The evaluations use different channel sets and
57 time windows and are reported separately.

58 Here, we audit source-only model selection as part of cross-subject decoder evaluation [11]. The
59 development analyses trace how training-duration decisions affect apparent model failure and
60 distinguish concentrated aggregate gains from participant-level benefit. Separate frozen external
61 evaluations retain adverse and uncertain outcomes, with participants as the unit of inference. A

62 downstream BNCI sensor analysis adds measured-signal context while keeping physiological
63 description separate from decoder attribution.

64 **Related Work**

65 Classical motor-imagery decoders use spatial and spectral structure. CSP estimates discriminative
66 spatial projections from class covariance structure [12], and filter-bank CSP combines spatial
67 features across frequency bands [13,14]. These methods inform the spatial and spectral controls
68 used here. The implemented filter bank is FBCSP-inspired rather than an exact reproduction; its
69 projections, feature scaling and classifiers are fitted on source partitions under the same target-
70 information restriction as the neural models.

71 EEGNet combines temporal filtering, depthwise spatial filtering and separable convolutions in a
72 compact architecture [15]; broader CNN work investigates learned EEG decoding and
73 visualization [16]. Our shared-input model processes predefined mu and beta views with one
74 configured EEGNet and averages their class logits. Its parameter count matches one branch, but
75 computation on two views and shared BatchNorm exposure differ. These bands define
76 experimental input views; they do not provide verified physiological source separation.

77 Algorithm reviews and multi-dataset benchmarks document performance variation between
78 methods, people and datasets [3,4]. Domain-generalization evaluation must also account for model
79 selection [11]. Here, fitting is restricted to source participants within a repeatedly used
80 development benchmark, followed by external evaluation of frozen pipelines. Methods that fit on
81 target labels or unlabeled target distributions have a different information budget and were not
82 compared directly.

83 Recent domain-generalization studies seek invariance across source domains for unseen-subject
84 motor-imagery decoding. EEG-DG aligns marginal and conditional distributions across multiple
85 source domains [17]. Zheng et al. combine spectral/task knowledge distillation with cross-source
86 correlation alignment and distance regularization [18]. These approaches pursue source-only
87 generalization without target adaptation, although a matched comparison still requires compatible

88 information budgets and selection procedures. Our contribution is an audit of source-only model
89 selection, participant heterogeneity and complete frozen-pipeline transfer, with adverse and null
90 results retained in a traceable evidence record. We do not introduce a new domain-generalization
91 algorithm or rank the evaluated pipelines against these methods.

92 EEG pretraining offers another route to transferable representations. LaBraM learns generic
93 representations from large EEG collections [19], while EEGPT uses pretrained transformers for
94 transferable EEG representations [20]. Fair comparison requires the pretraining data, downstream
95 fitting procedures and computational budgets to be specified. Neither model was evaluated here,
96 and equivalence to our target-information contract was not established; the findings concern the
97 selected source-only pipelines.

98 Event-related desynchronization and synchronization describe decreases and increases in spectral
99 power relative to a reference interval [21]. Earlier motor-imagery work investigated sensorimotor
100 rhythm changes [22]. These concepts guide the separate characterization of measured signals, for
101 which the baseline and estimator are specified explicitly. Scalp power patterns do not establish
102 anatomical source localization or demonstrate that a learned decoder used those patterns.

103 **Materials and Methods**

104 **Experimental Design**

105 The study comprises exploratory development and two external-evaluation stages. In the four-class
106 BNCI2014_001 development stage, both sessions of each target person are excluded from the
107 corresponding fitting and duration selection. Earlier benchmark outcomes informed later
108 candidates, so the nine-person comparisons remain exploratory. The earlier PhysioNet external
109 evaluation (Q14) applies separately fitted binary BNCI models to PhysioNet; a metadata-driven
110 sampling-rate amendment followed partial target execution. The frozen Cho2017/Lee2019 transfer
111 experiment (Q15) uses a new binary source pipeline with a different montage, crop and reference,
112 evaluating its frozen broad, shared-input and CSP models on Cho2017 and Lee2019 offline-

113 training segments. These datasets are external to fitting, but participant identity nonoverlap across
114 providers was not independently established.

115 The primary external estimand is the mean paired balanced-accuracy difference between the
116 frozen shared-input and broad pipelines, with equal weight for each person and separate estimates
117 for each cohort. The PhysioNet evaluation uses 17 shared and 18 broad epochs; the frozen
118 Cho2017/Lee2019 evaluation uses 19 shared and 14 broad epochs, all selected on source data.
119 Each contrast therefore evaluates complete selected pipelines rather than the isolated effect of
120 sharing. CSP provides context, and neither external stage compares rank with mean-loss selection.
121 The descriptive BNCI physiological analysis (Q16) was defined after decoder results were known
122 and uses BNCI signals only. Its raw-signal summaries and associations provide no feedback into
123 the decoders and cannot establish their physiological mechanism.

124 The corresponding saved summary is presented in Fig 1.

125 Fig 1. Study workflow and source–target information boundaries. Study workflow and information
126 boundaries. Both sessions of each internal target person are excluded from the corresponding
127 source fit; these development analyses remain exploratory. Q14 and Q15 use separate binary
128 models and preprocessing contracts. At inference, deterministic adapters and frozen source-
129 derived transformations are applied without fitting, adaptation or BatchNorm updates. Event and
130 class metadata establish eligibility and canonical label mapping; ground-truth labels are used for
131 scoring. The separate BNCI-only physiological analysis reads native signals and labels after
132 decoder outcomes are known, without feeding back into fitting or selection. Results are aggregated
133 by person. Q14's post-partial-run amendment and the external differences in selected duration are
134 described in the Methods and comparison-fairness table.

135 **Datasets, Populations and Task Separation**

136 BNCI2014_001 corresponds to BCI Competition IV dataset 2a [1,2]. Nine participants completed
137 two recording sessions, each containing six runs of 48 trials: 12 each of left-hand, right-hand, feet,
138 and tongue imagery. Twenty-two EEG channels and three EOG channels were sampled at 250 Hz.

139 Acquisition used a left-mastoid reference, right-mastoid ground, 0.5–100 Hz band-pass, and 50-Hz
140 notch; the offline operations below were applied in addition to this acquisition processing. The
141 primary neural input contains EEG channels only. The archived metadata contain 5,184 trials, 576
142 per participant and 1,296 per class. All 488 expert artifact-flagged trials remain in the primary
143 evaluation population. Including all trials differs from the original competition’s artifact-free
144 scoring; source-clean conditions exclude flagged source-training trials but retain every target trial.
145 The external EEG Motor Movement/Imagery Database version 1.0.0 [5] provides a separate binary
146 task. We include all 109 subjects and only imagery runs 4, 8, and 12; T1 and T2 denote left- and
147 right-fist imagery. Executed-movement, bilateral-task, and rest events are excluded from this
148 contrast. The final audited inventory contains 327 EDF files verified against official checksums
149 and 4,918 unique trials. Although the acquisition description specifies 160-Hz sampling, the file
150 headers identify three subjects with 128-Hz recordings. Four-class and binary balanced accuracy
151 have different chance levels and are reported separately.

152 Cho2017 [7,8] provides left/right-hand imagery recordings from 52 people at a native sampling
153 rate of 512 Hz. The audited evaluation retains 10,520 imagery trials: 49 people have 200 trials and
154 three (S7, S9, S46) have 240. The Lee2019 motor-imagery data [9,10] contain 54 people recorded
155 at 1,000 Hz in two sessions. We retain only EEG_MI_train, the provider’s offline-training runs,
156 giving 100 trials per session and 10,800 trials in total. EEG_MI_test, other paradigms, and non-
157 imagery events are excluded. The provider’s term “train” identifies a recording segment; these
158 external trials were used only for inference and scoring, never for parameter fitting. The three
159 external cohorts cannot be assumed to represent 215 independently verified unique identities
160 across providers.

161 The corresponding saved summary is presented in Table 1.

162 Table 1. Distinct populations and computational endpoints.

Analysis	People	Classes	Unique trials	Primary tensor
Internal BNCI	9	4	5,184	22×750 at 250 Hz
Q14 binary BNCI source	9	2	2,592	22×480 at 160 Hz
Q14 external PhysioNet	109	2	4,918	22×480 at 160 Hz
Q15 binary BNCI source	9	2	2,592	21×320 at 160 Hz
Q15 external Cho2017	52	2	10,520	21×320 at 160 Hz
Q15 external Lee2019	54	2	10,800	21×320 at 160 Hz

163 Both sessions of each BNCI outer target are held out. Q14 and Q15 have separate binary source
164 fits and preprocessing contracts, and neural inputs contain EEG channels only. Prediction rows
165 repeat trials across models and seeds; they are not independent participants. Lee sessions are
166 pooled within person for the primary endpoint.

167 **Strict Zero-calibration Information Contract**

168 Within each evaluation fit, strict zero-calibration means that no parameter, state, preprocessing
169 coefficient, hyperparameter, duration, seed, decision threshold or model choice is fitted or selected
170 using the current target person’s recordings or labels. Neural weights and BatchNorm statistics,
171 shallow-model coefficients, source-fitted transforms and source-selected schedules are fixed
172 before target inference. Target data are not used to fit normalization, covariance alignment or
173 adaptation, update BatchNorm, stop training or select seeds. This restriction applies to each fit; it
174 does not make the repeatedly used BNCI development programme independent of earlier target
175 outcomes.

176 Target EEG provides model inputs; event and class metadata determine declared eligibility and
 177 canonical label mappings. Labels are then used to score saved predictions. Cho recordings are
 178 stored in class-specific containers, so the adapter has access to native class metadata. Fixed
 179 filtering, channel selection, resampling and common-average referencing operate on target signals
 180 without fitting a target-dependent decoder. The Q15 Helmert basis depends only on channel count;
 181 scalers, CSP/PCA, whitening and EOG-regression coefficients, where used, are estimated on
 182 source-training data and applied unchanged. Separate physiological summaries use raw signals and
 183 labels but provide no feedback into bands, crops, weights, thresholds, schedules or model choice.
 184 The absence of target fitting does not establish physical acquisition calibration. Q15's native-
 185 numeric-as-microvolt convention does not verify the exported voltage units. Earlier within-person
 186 and cross-session P2 models use target-person labelled calibration and remain outside the strict
 187 zero-calibration endpoints. The source-fitted P4 EOG regression requires synchronous target EOG
 188 and thus additional sensors, although it fits no target coefficients.
 189 The corresponding saved summary is presented in Table 2.
 190 Table 2. Target information and fitting boundaries.

Operation	Information read	Target fitting	Interpretation
Eligibility, fixed channel and class maps, cue indexing	Native rate/channel/event/class metadata	None	Declared metadata adapter
Filtering, resampling, crop and Q15 CAR	Target EEG, fixed native metadata	None	Deterministic offline operations
Q15 Helmert coordinates	Channel count and signal	None	Analytical CAR-subspace basis
Source-fitted scaler/CSP/PCA/whitenin g/EOG regression	Source data to fit; target signal to apply	None	Frozen source coefficients; EOG needs extra sensors

Operation	Information read	Target fitting	Interpretation
Neural inference	Target tensors	None; weights and BatchNorm fixed	Strict zero-calibration prediction
Scoring and separate Q16 characterization	Predictions/raw signals and canonical labels	None	Downstream analyses with no decoder feedback
P2 within-person or cross-session models	Target-person labeled training trials	Yes	Calibrated context; separate endpoint

191 The restriction applies to fitting and selecting the corresponding decoder. It permits deterministic
192 computation on target signals, metadata inspection, and label use for scoring and separate
193 physiology. These boundaries do not make the development history outcome-blind.

194 **Offline Preprocessing and Source-only Partitions**

195 For internal BNCI analyses, fourth-order zero-phase Butterworth filters are applied within each
196 native recording run before epoch extraction. The half-open interval [2.5,5.5) s after the MAT trial
197 start corresponds to [0.5,3.5) s after the cue and yields 750 samples. The primary neural pipeline
198 applies no baseline subtraction, additional notch, or additional re-reference. MNE-loaded values in
199 volts are multiplied by 10^6 to give network inputs in microvolts. These offline zero-phase
200 operations use future samples, so this pipeline does not constitute a causal online decoder.

201 The broad mean-loss and mean-rank EEGNet conditions use 4–40 Hz input. Restricted broad input
202 uses 8–30 Hz, and the shared representation uses fixed mu (8–13 Hz) and beta (13–30 Hz) views.
203 These predefined bands are experimental choices; they are neither participant-specific estimates
204 nor physiological source separation. Any fitted scaler, CSP/PCA projection, or whitening
205 transform is estimated only on the applicable source-training partition.

206 Each primary four-class outer split holds out both sessions of one participant. The eight source
207 people contribute 4,608 trials and the target contributes 576. Sorted source IDs define four
208 consecutive, two-person inner-validation groups. Each inner fit trains on six people/3,456 trials
209 and validates on two people/1,152 trials. Thus, source training and validation are separated by

210 person. Final evaluation uses three fixed seeds, with scores averaged within person before
211 aggregation across people. BatchNorm statistics remain fixed during model evaluation and are not
212 updated by target examples.

213 **Training-duration Selection and Diagnostic Controls**

214 The mean-loss rule selects the earliest epoch that minimizes validation cross-entropy averaged
215 with equal weight across the four inner folds. Candidate epochs are 1–40. The mean-rank rule
216 ranks epochs within each fold by validation cross-entropy, assigns average ranks to ties, averages
217 the ranks across folds, and selects the earliest minimum. This rule is invariant to strictly increasing
218 (order-preserving) transformations of the losses within each fold, although that property does not
219 guarantee better target prediction. The rank follow-up reuses the 36 frozen original inner
220 trajectories and trains 27 new final models after the schedule has been frozen.

221 Selected epoch = earliest argmin over e of $(1/K) \sum_k \text{rank}_k[L_k(e)]$.

222 The rank candidate was motivated by earlier results on the same BNCI participants, so the internal
223 comparison remains exploratory despite source-only selection within each fit. A four-cell
224 diagnostic for S3 crosses raw and source-normalized input with two and 16 fixed epochs. Here,
225 raw input means filtered, microvolt-scaled EEG without fitted source normalization. Two cells
226 reuse the archived original conditions and two are new diagnostic conditions preserved in the
227 archive; the comparison involves one participant and does not provide population-level replication.

228 Source normalization estimates per-channel mean and scale from source-training samples only.

229 Later controls compare fixed 20-epoch training with raw-loss-selected schedules. A matched-

230 runtime experiment retains the historical source-selected epoch counts but retrains the 27 final

231 models using the same runtime and raw files as the fixed-duration comparator. Epochs are not

232 reselected under the newer runtime. Source-count controls use $k = 2, 4$, or 6 source participants,

233 four predefined identity subsets per count, and the $k = 8$ anchor; all counts use fixed 20 epochs.

234 The 0train-versus-1test source-session contrast has equal training budgets: 2,304 trials, 20 epochs,

235 and 720 optimizer updates per fit. Source-count contrasts and comparisons of one source session

236 with the both-session anchor change both the number of training examples and optimizer updates,
237 preventing attribution to a single causal factor.

238 The population raw-versus-source-normalized Q5/Q6 comparison also changes the selected
239 duration for S3 from 2 to 16 epochs and S5 from 13 to 12. Its aggregate difference therefore
240 cannot be attributed solely to normalization. The fixed-duration Q7 four-cell diagnostic separates
241 normalization and duration more directly, but only for S3.

242 **Neural Representation and Baseline Fairness**

243 The four-class model is Braindecode 1.5.1 EEGNet with 22 input channels, 750 samples, $F1 = 8$,
244 depth multiplier $D = 2$, $F2 = 16$, temporal kernel length 64 samples, and dropout 0.25. It has 2,932
245 trainable parameters in the archived implementation. Training uses Adam, learning rate 0.001,
246 batch size 64, cross-entropy, zero weight decay, and no scheduler or default augmentation. The
247 selection seed is 20260923 and the three final seeds are 20260924, 20260925, and 20260926. This
248 is a configured EEGNet study, not an exact reproduction of the original paper's preprocessing or
249 splits [15].

250 For shared mu/beta input, both band views pass through the same EEGNet in one concatenated
251 mini-batch. Their class logits are averaged before cross-entropy during training and before softmax
252 during inference. Network parameters and BatchNorm are shared, while dropout acts per view.

253 The model has no learned band weights, attention mechanism, or independently trained branches.

254 Its parameter count matches one EEGNet, but forward-pass computation and preprocessing differ.

255 Shared prediction: $p = \text{softmax}[(f\theta(X\mu) + f\theta(X\beta))/2]$.

256 Controls include individual frequency bands; fixed source-selected durations; Welch log-
257 bandpower with source-fitted shallow classifiers; eight source-only CSP or PCA time-series
258 projections per band; log-Euclidean covariance features; source-only normalization; and source-
259 clean training. The CSP/PCA neural controls both use 16 projected channels and source-fitted
260 standardization. Their input widths therefore match each other but differ from the 22-channel

shared-logit model. Architectural ablations use two independent networks, early band stacking, four shared bands, and a broadband capacity comparator. The independent two-band model has 5,864 parameters and its comparator has 5,914, a 0.85% mismatch. The traditional filter-bank implementation is FBCSP-inspired rather than a literal reproduction of Ang et al. [13,14].

The corresponding saved summary is presented in Table 3.

Table 3. Matching and remaining differences in decoder comparisons.

Comparison	Matched aspects	Remaining difference / scope
Q5 mean-loss vs Q8 mean-rank	Nine-person all-trial LOSO, 22×750, 4–40 Hz, original 36 inner trajectories, final seeds	Schedule changes; candidate arose after BNCI outcomes; exploratory
Q5 raw vs Q6 source normalization	Same all-trial targets and EEGNet family; source-only scaler fit	Selected epochs also change: S3 2 → 16, S5 13 → 12; not normalization alone
Q7 S3 raw/normalized ×2/16 epochs	Epochs fixed within each raw/normalized pair	One-person diagnostic
Q9 broad vs shared, reselected durations	Same targets, dimensions, EEGNet parameter count, optimizer/seeds	Bands, two-view computation, BatchNorm exposure and schedules differ
Q9-E002 fixed-schedule broad/shared	Same targets and frozen Q8 per-target epochs	Duration-controlled sensitivity; views/computation still differ
Q10 CSP vs PCA neural projections	Eight source-fitted components per band; 16 projected channels and source scaling	Matched to each other, not to 22-channel shared-logit input
Q11 independent branches vs capacity comparator	Same targets; approximate trainable capacity	5,864 vs 5,914 parameters; branch computation/normalization differ
Q12 GroupDRO vs balanced ERM	Same runtime and batching/sampling policy	Declared objective comparison

Comparison	Matched aspects	Remaining difference / scope
Q12 whitening/augmentation vs historical Q8	Same nominal target population and source-only rule	Historical baseline uses another runtime
Q13 fixed20 vs matched-runtime CE schedule	Same actual runtime, raw files and three seeds	Historical epochs replayed, not reselected in the newer runtime
Q13 source-session 0train vs 1test	2,304 source trials, 20 epochs, 720 updates	Session composition changes; both-session anchor also changes exposure
Q14 external broad vs shared	109 people/4,918 trials; 22×480; same source seeds and target rule	18 vs 17 epochs; metadata amendment after partial execution
Q15 external broad vs shared	Identical people/trials within cohort; 21×320 adapter, neural family/seeds	14 vs 19 epochs and one-view vs two-view processing
Q14/Q15 CSP4+LDA context	Same cohort scoring population; source-only fit	One shallow model vs three neural seeds; Q15 Helmert; differs from Q4 CSP8/OAS

Each comparison controls the aspects listed, while retaining the stated implementation differences. Q14/Q15 therefore compare complete frozen pipelines. Matching selected aspects does not isolate every causal factor, and the CSP comparisons remain descriptive.

Source-only Robustness Interventions

The six completed robustness conditions are source-pooled whitening, source-balanced empirical risk minimization (ERM), source-group distributionally robust optimization (GroupDRO), channel dropout, gain perturbation, and their combination. Source whitening uses equal-person covariance aggregation, a ridge of 5% of the mean eigenvalue, and scale preservation; no target covariance or alignment is fitted. GroupDRO treats each source person as a group [23]. Persistent log weights increase by 0.05 times detached per-group cross-entropy, are normalized by log-sum-exp, and weight the group loss. Its matched comparator is balanced ERM, which preserves the batching/sampling intervention. Training-only channel dropout uses probability 0.1 with

279 surviving-channel scale 1/0.9; gain perturbation uses $U[0.8,1.2]$ and precedes dropout in the
280 combined condition. Validation and target inputs are not augmented.

281 Whitening and augmentation conditions are compared with the previously frozen broad rank-
282 selected baseline. The historical baseline ran on a different runtime from the robustness batch, so
283 these differences reflect both intervention and runtime variation. GroupDRO versus balanced
284 ERM provides a within-runtime comparison of training objectives. This distinction accompanies
285 the numerical table and limits attribution of cross-batch differences to the named interventions.

286 **Earlier Binary Source Freeze and PhysioNet Evaluation (Q14)**

287 The binary BNCI source population contains 2,592 left/right trials, 288 per person. Filtering and
288 epoching occur at 250 Hz before deterministic polyphase resampling by 16/25 to 160 Hz, giving
289 22×480 samples. The broad binary model uses 8–30 Hz; the shared model uses the fixed mu/beta
290 views. Source-only duration selection across all nine participants uses validation groups
291 comprising people 1–2, 3–4, 5–6, and 7–9, with equal fold weighting. The frozen training
292 durations are 18 epochs for the broad model and 17 for the shared model. Checkpoints, channels,
293 labels, contrast, seeds, and preprocessing are fixed before external access. The source-only CSP4 +
294 LDA comparator uses $\text{reg} = \text{None}$ and LDA solver = svd; it differs from the internal CSP8
295 OAS/shrinkage-LDA configuration. The external neural comparison evaluates complete pipelines
296 with different source-selected durations and therefore cannot isolate parameter sharing.

297 A partial external run stopped at S88 because its 128-Hz header violated the original 160-Hz
298 assumption. A versioned amendment used metadata from the remaining headers without
299 consulting predictions or scores. S88, S92, and S100 are filtered and epoched at their native rate,
300 then resampled by 5/4 from 384 to 480 samples. The other 106 subjects retain native 160-Hz
301 processing. The 87 completed subject outputs from the earlier run are copied byte-for-byte, and 22
302 additional subjects complete the cohort. The amendment introduces no target statistics, fitted
303 normalization, adaptation, seed selection, or class-performance-dependent processing. Because it

304 followed partial external execution, the combined result does not represent an unmodified
305 execution of the original freeze.

306 The original aggregate validation failed because it applied an inappropriate exact floating-point
307 equality check to re-serialized CSV probabilities. A separate cross-host runtime mismatch later
308 prompted a versioned portability amendment. Both limitations are documented. The versioned
309 validation-only audit checked official EDF checksums and event identities and recomputed saved
310 probability/argmax consistency, subject metrics, and statistics. It passed with zero new fits and
311 zero new model-inference rows. It therefore validates saved outputs; it does not regenerate all
312 external predictions from checkpoints on the later validation host.

313 **Frozen Cho2017/Lee2019 Transfer (Q15)**

314 The frozen Cho2017/Lee2019 study is a separate binary transfer experiment using different
315 checkpoints from Q14. All nine BNCI participants supply 2,592 left/right trials. The common
316 ordered montage comprises Fz, FC3, FC1, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz,
317 CP2, CP4, P1, Pz, P2, and POz. Each trial is processed in a fixed native-rate context from -1.5 to
318 $+4.5$ s relative to the cue. Common-average reference is computed across these 21 selected
319 channels, followed by native-rate zero-phase second-order-section Butterworth filtering with
320 prototype order four. Broad input uses 8–30 Hz; the shared views use 8–13 and 13–30 Hz.

321 Deterministic polyphase resampling yields 160 Hz, followed by the half-open cue crop $[0.5, 2.5)$ s,
322 equivalent to samples $[320, 640)$ of the resampled six-second context. The resulting model tensor is
323 21×320 . These preprocessing and input differences preclude interpreting a comparison with Q14
324 as an effect of external provider alone.

325 The Q15 adapter interprets native numeric values as microvolts by convention. Physical voltage
326 calibration, the original Cho hardware reference, and hardware cue latency have not been
327 independently verified. The first 64 of the Cho payload's 68 rows are EEG; four EMG rows are
328 excluded. Channel mapping follows the publication's numbered montage because the payload
329 does not supply names. Cho native labels left = 1 and right = 2 are retained. Lee native labels left =

2 and right = 1 are deterministically mapped to canonical left = 1 and right = 2. Lee MATLAB cue indices are converted using $t - 1$; the provider's stored segments align with Python t , leaving a disclosed one-sample (1-ms) indexing discrepancy that was not changed using outcome scores. The supplement documents these adapter assumptions and the file-level checks.

Source-only duration selection uses four groups comprising people 1–2, 3–4, 5–6, and 7–9, candidate epochs 1–40, equally weighted mean within-fold cross-entropy ranks, and the earliest minimum. Final durations are 14 broad epochs and 19 shared epochs. The selection seed is 20260923, and final seeds are 20260924–20260926. Training uses the same configured EEGNet family, optimizer, and fixed-band logit averaging described above, with the Q15 channel/time dimensions. A source-fitted CSP4 + LDA comparator applies a fixed 21-to-20 Helmert coordinate transform after common-average reference, followed by source-only fitting. The external neural comparison evaluates two complete frozen pipelines with different source-selected durations and cannot isolate parameter sharing or band decomposition alone. Source training comprises eight inner neural fits, six final neural fits, and one shallow fit: 15 in total. Migration reused these artifacts without additional source or target fitting.

Official raw metadata for 18 BNCI files and all declared Cho/Lee files were audited before the preprocessing specification was committed; its pre-fit publication was recorded at 2026-10-04 01:33:03 UTC. Source training was completed at 01:52:42 UTC. The separate inference freeze was committed on 2026-10-05 at 09:31:05 UTC, before external prediction, fixing channels, timing, label adapters, transformations, checkpoints, seeds, scoring units, and primary statistical procedures. Event metadata may be inspected to determine preprocessing eligibility, but external labels are not used for fitting or model selection. BatchNorm remains in evaluation mode; there is no target alignment, normalization fit, adaptation, or post hoc seed selection. These repository-defined freezes followed external metadata inspection and do not constitute independently registered preregistration.

355 All 52 Cho2017 and 54 Lee2019 participants were processed under the declared adapters. The
356 archived independent computational validator replays processing from raw data to epochs and
357 predictions from the frozen models, checks canonical events and scoring populations, verifies links
358 to inference and source artifacts, and recalculates aggregate statistics. Validation passed subject to
359 the stated acquisition-calibration limitations. The supplementary reproducibility documentation
360 identifies the exact input, parameter, checkpoint and result revisions.

361 **BNCI-only Physiological Signal Characterization (Q16)**

362 The BNCI characterization separately describes native BNCI2014_001 recordings: 18 files, 108
363 motor-imagery runs, nine people and both sessions, retaining all 5,184 four-class trials and their
364 artifact flags in the primary population. The signals retain all 22 EEG sensors at native 250 Hz,
365 with no additional bandpass, notch, resampling, reference transform, source scaler or learned
366 projection. Three EOG channels are excluded. EEG column order follows the provider montage;
367 individual MAT files do not contain channel-name strings. Native one-based trial starts are
368 converted to zero-based samples; the cue is 500 samples after the trial start. The fixed cue-relative
369 baseline is $[-1.5, -0.5]$ s and the task window is $[0.5, 2.5]$ s. Before power calculation, all originals
370 are rehashed against archived source provenance, and labels, artifact flags, finite samples, montage
371 order and complete window bounds are checked. All 18 source SHA-256 digests, 5,184 event
372 identities and 228,096 trial-channel-band eligibility rows passed these checks; no trial was
373 excluded by a power guard. The primary population retained all 488 flagged trials, including 246
374 of the 2,592 hand trials.

375 Welch power is defined separately within each trial's baseline and task windows using float64
376 native numeric values, a periodic Hann taper, 250-sample segments, 125-sample overlap, 250-
377 point FFT, linear detrending, one-sided density and arithmetic segment averaging. This gives 1-Hz
378 bins. Closed 8–13-Hz and 13–30-Hz masks are integrated with the trapezoidal rule in physical
379 frequency; the 13-Hz boundary supplies half-bin weights to the adjacent integrals. The trial-level
380 change is $10 \log_{10}(P_{\text{task}}/P_{\text{baseline}})$, in dB. Nonfinite or nonpositive powers require an explicit

381 invalid-row receipt rather than an epsilon or clipped ratio. The 1-s baseline contributes one Welch
 382 segment and the 2-s task contributes three overlapping segments. Unequal log-power estimator
 383 variance can create a positive offset even without a true power change, so these dB summaries are
 384 not claimed to be unbiased ERD estimates.

385 Arithmetic mean trial dB is aggregated within person, session, class, channel and band, followed
 386 by equal averaging of the two sessions. This averages log ratios rather than taking the logarithm of
 387 averaged powers. Left/right-hand C3/C4 summaries use common trial eligibility. The signed hand
 388 descriptor is $L = [(C3_{\text{right}} - C4_{\text{right}}) + (C4_{\text{left}} - C3_{\text{left}})]/2$. A negative descriptor means the
 389 contralateral task/baseline change is lower than the ipsilateral change; it does not alone establish
 390 absolute contralateral ERD. C3, Cz and C4 absolute changes, separate hand terms, full scalp
 391 summaries, sessions and artifact strata are retained to show that distinction.

392 The physiological parameters were fixed on 6 October 2026 after the decoder outcomes were
 393 known and before the physiological calculations. The analysis is therefore descriptive and does not
 394 provide independent confirmation. Associations use existing matched-binary Q14-E001 BNCI
 395 LOSO scores, averaged over the three fixed neural seeds within each of the same nine people; all-
 396 nine-source fits and external-cohort scores are excluded. The broad EEGNet score is the
 397 designated descriptive endpoint. Six Spearman coefficients cover the three fixed models and two
 398 fixed bands, with no p-values, tertile groups or outcome-selected bands/models. Decoder imagery
 399 covers [0.5,3.5) s, whereas the physiological power analysis covers [0.5,2.5) s. No external
 400 physiology was calculated, and no physiological output was used to fit, select or update a decoder.
 401 The parameter freeze is commit 050e01b028aaab8e3d745934b13b2d17e9bb0a7a; its public
 402 readback preceded raw-power calculation. The preprocessing freeze, raw-data receipt, execution
 403 manifest and final independent validation link the specification and inputs to the completed tables
 404 in Q16-P001-BNCI-20261006.

405 **Statistical Analysis**

406 $\text{Balanced accuracy} = (1/C) \sum_c \text{TP}_c / (\text{TP}_c + \text{FN}_c).$

407 The held-out person is the unit of statistical inference. Paired seeds are averaged within person;
408 predefined source-identity subsets are averaged within person and source count. Shallow
409 conditions have one prediction per trial rather than three independent neural seeds. Paired
410 differences are expressed in balanced-accuracy percentage points (pp). Person-bootstrap percentile
411 intervals resample people, not trials, sessions, seeds, or source subsets. External intervals describe
412 variation across target people conditional on the frozen source models; they do not quantify
413 source-cohort sampling uncertainty. Internal analyses remain exploratory because benchmark
414 outcomes informed the research program and outer training sets overlap. For Lee2019, the two
415 retained sessions are pooled before calculating each person/seed balanced accuracy. The three
416 neural-seed scores are then averaged within person, giving an average of scores rather than an
417 ensemble of probabilities. A single CSP score is retained per person. Cross-cohort score
418 differences are descriptive because Q14 and Q15 differ in source preprocessing and checkpoints.
419 For each archived experiment, we report its original statistics and the interval resampling
420 procedure. Rank-selection intervals use 20,000 person-bootstrap draws with seed 20260923. The
421 archived shared-minus-broad representation contrast and robustness intervals use 20,000 draws
422 with seed 20260924. Source-composition and matched-runtime intervals use 10,000 draws with
423 seed 20260926 and exhaustive two-sided sign flips over nine participant differences; familywise
424 exploratory p-values use Holm correction within the frozen families. The earlier PhysioNet
425 primary contrast uses 20,000 draws with seed 20260924 and an exact two-sided binomial sign test
426 excluding exact zero differences. Its primary hypothesis is shared-minus-broad; contextual
427 comparisons with CSP do not replace it. A supplementary numerical audit uses 200,000 draws
428 with seed 20261002 and is labelled separately. An interval excluding zero need not agree with
429 every small-sample test. For Q15, the predeclared family contains exactly two shared-minus-broad
430 contrasts, one for each external cohort. Percentile person-bootstrap intervals use 20,000 draws with
431 seed 20260924 [24]. Two-sided person-level sign-flip tests use 20,000 Monte Carlo draws with
432 seed 20261003, assuming sign exchangeability of paired person effects under the null, evaluating

433 the absolute mean, and applying the $(\text{exceedances} + 1)/(20,000 + 1)$ correction [25]. Holm
434 adjustment controls this two-contrast family [26]; it does not retroactively include the earlier
435 PhysioNet or internal hypotheses. The smallest attainable Monte Carlo p-value is $1/20,001$. The
436 intervals are unadjusted 95% intervals, and CSP comparisons are descriptive context. Descriptive
437 Q15 counts of positive, negative, and tied person effects use an absolute 10^{-15} tolerance for ties;
438 the frozen primary bootstrap and sign-flip calculations retain the saved numerical effects.
439 One historical rank-selection script labelled a nine-person binomial calculation as a sign test while
440 retaining two unchanged participants in its denominator. We report its $p = 0.1797$ as an archived
441 nine-person sign sensitivity. A conventional tie-excluding calculation uses seven nonzero changes
442 and gives $p = 0.015625$; the paired t-test gives $p = 0.111$. These procedures are reported separately
443 rather than treated as interchangeable or used to select a favourable result. Multiplicity corrections
444 from later declared families are not retrospectively applied to unrelated earlier statistics.
445 The physiological characterization was specified after decoder outcomes were known and remains
446 descriptive. Physiological aggregation and decoder associations use nine person-level records;
447 channels, trials, sessions and seeds are not treated as independent people. The six Spearman
448 coefficients have no physiological p-values or confirmatory testing family. They are separate from
449 Q15's frozen two-cohort tests and cannot retrospectively confirm the decoder contrasts.

450 **Reproducibility and Validation Scope**

451 Saved predictions were independently reaggregated to check trial identities, scoring populations,
452 person-level confusion matrices, seed-averaged scores and reported statistics. This numerical
453 reanalysis neither loaded decoder checkpoints nor generated predictions. The Cho2017/Lee2019
454 experiment also underwent a completed replay from raw data to epochs and frozen-model
455 predictions; the earlier PhysioNet portability audit checked saved outputs without replaying
456 inference. Validation of earlier experiments remains experiment-specific, as documented in the
457 supplement.

458 The physiological analysis was checked independently against native BNCI recordings. An
459 explicit detrending, periodic-Hann FFT and trapezoidal integration calculation reproduced all
460 31,104 C3/Cz/C4 trial-band pairs: the maximum relative power difference was 1.85×10^{-15} and
461 the maximum absolute dB difference was 7.11×10^{-15} , within fixed 10^{-10} tolerances. Raw
462 spectral replay covered these three central channels; identities and eligibility were checked for all
463 228,096 saved rows, and all aggregation and association tables were recomputed. Numerical
464 agreement establishes implementation consistency, not independent laboratory replication.
465 OpenAI Codex assisted analysis-code development and verification-code preparation. The
466 computations use explicitly specified algorithms; the archived validation reports document the
467 numerical checks. AI-assisted code generation is distinct from a learned decoder or a physiological
468 model.

469 **Ethics statement**

470 This study involved secondary analysis of public EEG recordings and recruited no new
471 participants. The Cho2017 and Lee2019 source publications report collection ethics review and
472 written informed consent [7,9]. Those collection approvals are not approvals for this secondary
473 analysis. The author confirms that neither ethics approval nor an exemption was required for this
474 secondary analysis. No formal approval or exemption documentation was obtained for this
475 secondary analysis; this statement records the author's assessment rather than an institutional or
476 committee determination.

477 The retained research records document access to BNCI2014_001 recordings from 21 September
478 to 6 October 2026, to PhysioNet EEG Motor Movement/Imagery recordings on 25 September
479 2026, and raw-data auditing and external evaluation of Cho2017 and Lee2019 MI from 3 to 5
480 October 2026. Dates refer to the recorded research-access periods in UTC; the calendar dates are
481 the same in China Standard Time. They do not establish first-ever download dates or exclude
482 access outside the retained records. Git commit, publication and backup timestamps were not
483 substituted for recorded data-use events.

484 The author reports access only to the recordings, provider-assigned participant codes and basic
485 demographic metadata, such as age, without names, contact details or an identity linkage key. No
486 direct identifying information was available to the author; public participant codes were retained
487 for within-dataset grouping and scoring.

488 **Data and code availability**

489 Public BNCI2014_001, PhysioNet, Cho2017 and Lee2019 recordings are available from the
490 original providers [1,2,5,6,7,8,9,10], subject to their licenses and terms of use. PhysioNet platform
491 publications are cited separately [27,28]. Analysis code, saved predictions, participant-level
492 summaries, parameter specifications and numerical validation reports are available at
493 <https://github.com/jackzhu119/cross-subject-mi-eeg>. The supplementary reproducibility
494 documentation identifies the exact source and result revisions. Raw EEG is not redistributed in the
495 manuscript bundle.

496 The current submission materials and licensing notices are available at
497 <https://github.com/jackzhu119/cross-subject-mi-eeg/tree/plos-one-submission-v1.1>. Author-owned
498 original software is licensed under MIT; author-owned original manuscript material and derived
499 outputs are licensed under CC BY 4.0. Third-party recordings, software, fonts and other
500 incorporated materials retain their original terms. These permissions do not alter frozen scientific
501 results or grant third-party rights.

502 **AI assistance and verification**

503 ChatGPT (GPT-6, as reported by the author) assisted information retrieval, grammar revision, and
504 preparation of instructions for publishing research materials to GitHub. OpenAI Codex assisted
505 source and literature checks, analysis-code and verification-code development, saved-result
506 consistency checks, figure preparation, and manuscript drafting and language revision. The author
507 personally reviewed and revised the final manuscript and takes responsibility for its scientific
508 content.

509 The underlying Codex model version is not reliably documented in the project records. Saved-
510 result checks compared reported numbers with frozen outputs and archived independent validation
511 reports; these checks did not rerun models or physiology experiments during manuscript
512 preparation. The author reports personal review and revision of the scientific content. No AI tool is
513 an author.

514 **Results**

515 **Traditional baselines and heterogeneous duration selection**

516 The all-trial four-class broad CSP8 + LDA baseline achieved 39.04% balanced accuracy; broad
517 CSP + SVM and the full filter-bank LDA comparator achieved 37.50% and 38.12%. EEGNet with
518 duration selected by raw mean loss achieved 33.72%, below the broad CSP baseline. These
519 descriptive comparisons do not establish superiority of a model class. The mean-loss schedule
520 selected only one or two epochs for S2, S3, and S8, while source-validation optima varied
521 substantially across folds.

522 Source-only per-channel normalization increased mean balanced accuracy to 37.93%, a +4.21 pp
523 difference. The archived paired 95% interval spans -1.12 to $+13.49$ pp, whereas the median
524 change was only $+0.23$ pp. Effects varied across participants, including negative transfer in S8, so
525 the average gain does not establish a general normalization benefit. Selected durations also
526 changed for S3 (2 to 16 epochs) and S5 (13 to 12); the population contrast therefore combines
527 normalization and duration changes.

528 The S3 diagnostic separates normalization from training duration. With two epochs, raw input
529 gave 26.10% balanced accuracy and source-normalized input gave 25.69%. With 16 epochs, raw
530 and normalized inputs gave 67.71% and 65.86%. The short-run raw condition assigned
531 approximately 94% of predictions to one class. Extending training reproduced the recovery for this
532 participant, but this four-cell diagnostic does not establish that duration explains all cross-person
533 failures.

Mean-rank selection increased equal-person mean balanced accuracy to 42.67%, a +8.95 pp change over raw mean-loss selection. The archived bootstrap interval is [+1.05,+19.96] pp, and the median change is +1.85 pp. Seven participants improved and two were unchanged. S3 and S8 account for 87.57% of the summed improvement. The average gain therefore reflects a concentrated benefit; prior development on the same dataset and disagreement among small-sample statistics limit its generalizability.

The corresponding saved summary is presented in Table 4.

Table 4. Selected completed four-class conditions; all nine participants.

Condition	Bands (Hz)	Mean BA (%)	Interpretation
Broad CSP8 + LDA	4–40	39.04	Single source fit/fold
Filter-bank CSP + LDA	4–40 bank	38.12	Fixed filter-bank control
EEGNet, mean loss	4–40	33.72	Source-selected duration
EEGNet, source norm	4–40	37.93	Source-only fitted scale
EEGNet, mean rank	4–40	42.67	Reused source curves
Shared mu/beta	8–13 / 13–30	42.19	Representation-specific rank
Broad, fixed 20	4–40	43.99	Later runtime
Broad, matched CE schedule	4–40	33.92	Same runtime as fixed 20

The primary population includes all trials. Neural scores are averaged over three fixed seeds within person before population summaries. Rows span different experimental batches, so runtime variation is controlled only in explicitly matched comparisons. The table summarizes declared controls rather than a target-selected ranking; full representation and robustness inventories are in the supplement.

The corresponding saved summary is presented in Fig 2.

548 Fig 2. Training-duration selection and participant-level internal results. A: archived source-
549 selected epochs. B: each participant's all-trial balanced accuracy averaged over three fixed seeds;
550 the dotted line marks four-class chance (25%). C: the S3 four-cell diagnostic, comparing raw and
551 source-normalized input at two fixed durations. Error bars show standard deviations across three
552 seeds, not uncertainty across people. This diagnostic concerns one person.

553 **Matched-runtime duration and source-composition controls**

554 In the additive matched-runtime experiment, the historical mean-loss duration schedule gave
555 33.92% balanced accuracy, compared with 43.99% for fixed 20 epochs. The fixed-minus-
556 historical-schedule mean was +10.06 pp, with a bootstrap 95% interval of [+0.68,+21.53] pp and
557 exact sign-flip $p = 0.125$. Four of nine people improved and five worsened; the median difference
558 was -0.174 pp. The positive mean thus reflects large recoveries rather than a typical-person
559 benefit. The average schedule difference persisted under matched runtime, but it does not establish
560 20 epochs as universally preferable: historical epoch counts were reused rather than newly selected
561 under that runtime.

562 Mean balanced accuracy increased descriptively with source count. Broad fixed-duration models
563 gave 32.96%, 38.89%, 43.16%, and 43.99% at $k = 2, 4, 6$, and 8; shared models gave 34.74%,
564 38.89%, 42.48%, and 43.98%. Performance varied substantially with source identity at each count.
565 The two-source broad-minus-eight-source difference was -11.03 pp, but its Holm-adjusted
566 exploratory p was 0.164. No source-count/selection/session family showed a corrected significant
567 result. Contrasts between the two sessions were small and heterogeneous. These trajectories are
568 consistent with sensitivity to source training information, but do not separate source diversity from
569 sample count or optimizer-update budget.

570 The corresponding saved summary is presented in Fig 3.

571 Fig 3. Source composition and matched-runtime controls. A: thick lines show equal-person means;
572 thin lines show individual target-person trajectories averaged over the four predefined identity
573 subsets and three seeds at $k = 2, 4, 6$. Every source count uses 20 epochs; $k = 8$ reuses the fixed-20

574 anchor. B: all nine fixed-20-minus-historical-CE-schedule effects in the matched runtime. The
575 interval reverses the sign of the archived matched contrast. Source count also changes data volume
576 and optimizer updates, so these exploratory sensitivities do not isolate source count alone.

577 **Spectral, spatial, and architectural interventions**

578 The internal shared mu/beta condition achieved 42.19% balanced accuracy, compared with
579 42.67% for the broad mean-rank baseline, a -0.48 pp mean difference. The archived Q10-V001
580 exploratory paired interval is $[-3.21, +1.41]$ pp (20,000 draws, seed 20260924), with exact sign-
581 flip $p = 0.914$. This comparison did not demonstrate a spectral-sharing gain. The restricted broad
582 band and individual bands provide descriptive frequency controls, while fixed-duration
583 representation controls retain their declared schedules.

584 The source-only CSP8 and PCA8 neural projections achieved 41.48% and 42.06% balanced
585 accuracy. Architectural controls achieved 42.99% for independent two-band networks, 42.19% for
586 the broadband capacity comparator, 42.66% for early stacking, and 39.80% for four-band sharing.
587 These alternatives provide no evidence of a general shared-band advantage. Differences in
588 preprocessing, projected width, BatchNorm behaviour, and compute also preclude attributing the
589 contrasts to a single physiological factor. Full model means and participant-level results
590 accompany the manuscript.

591 **Source-only robustness has no corrected established gain**

592 Source-pooled whitening, source-balanced ERM, and gain perturbation had positive point
593 estimates relative to the historical broad rank baseline, whereas dropout conditions were lower. No
594 intervention established an improvement after correction. The matched GroupDRO-minus-
595 balanced-ERM difference was -6.45 pp (unadjusted bootstrap interval $[-11.47, -1.79]$ pp; exact
596 sign-flip $p = 0.046875$, Holm-adjusted $p = 0.140625$). This adverse estimate warrants follow-up
597 but does not establish that distributionally robust optimization generally harms EEG decoding. The

598 whitening and gain perturbation intervals include zero, and comparison with a historical model
 599 from a different runtime further limits attribution.

600 The corresponding saved summary is presented in Table 5.

601 Table 5. Exploratory source-only robustness (nine participants).

Condition	BA (%)	Control	Δ (pp)	95% CI (pp)	Holm p
Whitening	44.34	Q8 historical	+1.67	[-0.39, +3.67]	0.3594
Balanced ERM	43.18	Q8 historical	+0.51	[-0.71, +1.77]	0.4766
GroupDRO	36.73	Balanced ERM	-6.45	[-11.47, -1.79]	0.1406
Channel dropout	41.00	Q8 historical	-1.67	[-4.51, +0.64]	0.7617
Gain perturbation	43.42	Q8 historical	+0.75	[-0.34, +1.86]	0.7617
Dropout + gain	40.92	Q8 historical	-1.75	[-5.05, +0.91]	0.7617

602 Intervals are the archived, unadjusted person-bootstrap intervals; p-values use the declared within-
 603 family Holm correction. Every comparison with Q8 crosses runtimes. GroupDRO versus balanced
 604 ERM matches batching within the same runtime. Neither intervals nor p-values are used to select a
 605 model on target outcomes.

606 **Earlier PhysioNet transfer: a small, uncertain primary effect**

607 In binary BNCI development, broad, shared, and CSP models achieved 68.21%, 67.35%, and
 608 61.54% balanced accuracy. These development outcomes cannot be pooled with the four-class
 609 analyses. The separately frozen all-nine-source models then generated predictions for all 109
 610 external subjects. Final validation confirmed 4,918 trial identities and 34,426 saved prediction
 611 rows. No model was fitted to target data.

612 External equal-person balanced accuracy was 61.81% for broad EEGNet, 62.39% for shared
 613 mu/beta, and 54.46% for CSP4 + LDA. The predeclared shared-minus-broad primary difference
 614 was +0.573 pp, with a 95% bootstrap interval of [-0.097, +1.242] pp. Sixty-three people favoured

615 shared, 43 favoured broad, and three tied; the exact tie-excluding sign-test p was 0.06446. The
616 interval includes zero, and the sign test does not establish external superiority.

617 Individual performance varied widely. The shared model ranged from 39.33% to 97.07% balanced
618 accuracy, with a median of 58.73%. Differences averaged across three seeds also varied in
619 direction across people. The three resampled subjects had mean shared-minus-broad $\Delta = +0.514$
620 pp, compared with +0.575 pp for the 106 native-160-Hz subjects. The three-person subgroup
621 comparison is descriptive and has no powered subgroup test. The result incorporates the sampling-
622 rate and validation-portability amendments and therefore does not represent an unmodified
623 execution of the original external protocol.

624 The corresponding saved summary is presented in Fig 4.

625 Fig 4. Separate-cohort binary transfer. A: empirical cumulative distributions of three-seed person-
626 mean balanced accuracy, with one CSP score per person; the dotted vertical line marks binary
627 chance (50%). B: all 109 paired primary effects, with zero and the mean indicated. The interval
628 and p-value are the frozen person-bootstrap and tie-excluding sign-test outputs. Three participants
629 were deterministically resampled under the disclosed metadata amendment.

630 The corresponding saved summary is presented in Table 6.

631 Table 6. Separate binary external evaluation (109 participants).

Frozen model	Mean BA (%)	Subject SD (pp)	Median BA (%)
Broad EEGNet	61.81	12.56	58.23
Shared mu/beta	62.39	12.92	58.73
CSP4 + LDA	54.46	7.90	52.08

632 Deep-model scores are averaged over seeds within person. The primary shared-minus-broad
633 contrast is +0.573 pp, 95% CI [-0.097,+1.242], p = 0.06446. CSP provides context and is outside
634 the primary hypothesis. The Methods describe the sampling-rate amendment and versioned
635 portability validation.

Completed Cho2017 and Lee2019 transfer: lower balanced accuracy on Cho2017 and an uncertain Lee2019 contrast

The completed Q15 outputs cover 21,320 unique external trials and 149,240 saved model/seed prediction rows, seven per trial. Independent checks verified canonical labels and event identities, person-level confusion matrices, aggregate balanced accuracies, frozen statistical outputs, and links to the declared artifacts. Original source fits remained 15, with zero new migration fits and zero target fits. These counts confirm source-only fitting; they do not constitute independent repetitions of the external experiment.

On Cho2017, equal-person balanced accuracy was 59.82% for broad EEGNet and 58.31% for shared mu/beta EEGNet. The primary shared-minus-broad mean was -1.513 pp (95% person-bootstrap interval $[-2.249, -0.804]$ pp), with two-sided Monte Carlo sign-flip $p = 0.000049998$ and two-cohort Holm-adjusted $p = 0.000099995$. The raw p -value is the Monte Carlo procedure's minimum attainable value and does not resolve probabilities below that value. Twelve people favoured shared, 37 favoured broad, and three were tied within a 10^{-15} numerical tolerance. Shared performed worse in this declared complete-pipeline comparison after correction.

On Lee2019, broad and shared models achieved 65.51% and 65.72% balanced accuracy. The primary difference was $+0.204$ pp (95% interval $[-0.515, +0.969]$ pp); raw and Holm-adjusted p -values were both 0.60367. Twenty-three people favoured shared, 30 favoured broad, and one tied. This small, uncertain contrast establishes neither a reliable improvement nor equivalence. The two sessions were combined within each person, so neither the 108 session records nor repeated model seeds increase the primary sample size beyond 54.

The corresponding saved summary is presented in Table 7.

658 Table 7. New external binary evaluation under the Q15 freeze.

Cohort / model	BA (%)	Person SD (pp)	Left recall (%)	Right recall (%)
Cho2017 / Broad EEGNet	59.82	8.64	46.79	72.86
Cho2017 / Shared mu/beta	58.31	7.22	37.52	79.10
Cho2017 / CSP4 + LDA	51.99	5.67	94.10	9.88
Lee2019 / Broad EEGNet	65.51	10.97	75.56	55.46
Lee2019 / Shared mu/beta	65.72	10.76	72.69	58.75
Lee2019 / CSP4 + LDA	52.77	5.57	85.98	19.56

659 Neural scores and class recalls are averaged over three seeds within person, then equally over
660 people. Person SD describes heterogeneity, not a standard error. Lee includes only offline-training
661 runs, pooled across two sessions. The primary shared-minus-broad contrasts are -1.513 pp
662 $[-2.249, -0.804]$, Holm $p = 0.000100$ for Cho, and $+0.204$ pp $[-0.515, +0.969]$, Holm $p = 0.604$ for
663 Lee. Class recalls and CSP comparisons are descriptive; no additional confirmatory tests were
664 added.

665 The CSP comparator was near binary chance on both cohorts and strongly favoured left-hand
666 labels. On Cho2017, mean left-hand recall fell from 46.79% with broad input to 37.52% with
667 shared input, while right-hand recall increased from 72.86% to 79.10%. On Lee2019, the
668 corresponding recalls changed from 75.56% to 72.69% and from 55.46% to 58.75%. Balanced
669 accuracy summarizes these asymmetric shifts but does not show which class gains or loses recall.
670 These descriptive checks used the frozen outputs and did not trigger any threshold, label, adapter,
671 or model change.

672 The corresponding saved summary is presented in Fig 5.

673 Fig 5. Frozen external evaluation on Cho2017 and Lee2019. Frozen Q15 external evaluation.

674 Cohorts are shown separately: person/model scores, equal-person means and person standard

675 deviations, and frozen paired person-bootstrap 95% intervals. Deep-model scores average three

676 seeds within person; Lee sessions are pooled before person scoring. Holm p-values refer to the

677 predeclared two-cohort shared-minus-broad family. The common 21-channel, two-second pipeline

678 differs from Q14, and no pooled cross-cohort test was specified.

679 **Descriptive BNCI Sensor Physiology and Decoder Associations (Q16)**

680 All 18 raw files, 108 labelled runs and 5,184 trials passed the fixed coverage checks. All 228,096

681 trial-channel-band rows had finite, positive baseline and task power. No trial, session or person

682 was excluded; the 488 provider-flagged trials remained in the primary summaries. Fig 6 shows

683 equal-person scalp means and all nine C3/Cz/C4 profiles. Task/baseline changes calculated with

684 the fixed estimator varied by sensor, imagined hand and person.

685 Fig 6. Descriptive native BNCI task/baseline power changes. A–D: equal-person hand-imagery

686 means for mu (8–13 Hz) and beta (13–30 Hz) at all 22 EEG sensors. In the schematic

687 standard-1020 geometry, anterior is upward and left scalp is on the viewer’s left; linear

688 interpolation is restricted to the sensor convex hull. No sensor or interpolated grid value is clipped

689 by the fixed ± 6 -dB range. E–F: all nine equal-session C3/Cz/C4 profiles (thin lines) and equal-

690 person means (thick lines). Trial dB uses cue baseline $[-1.5, -0.5]$ s and task $[0.5, 2.5]$ s, with one

691 versus three overlapping Welch segments. Negative values indicate lower task power under this

692 estimator; unequal window precision can bias log ratios. All trials and artifact flags are retained.

693 Template topography neither localizes cortical sources nor attributes decoder decisions.

694 Mu signed laterality averaged -0.250 dB, with six negative and three positive person descriptors;

695 beta averaged -0.168 dB, with eight negative and one positive descriptor (Table 8). Contralateral

696 and ipsilateral group means were both negative in each band, but absolute contralateral changes

697 were not negative for every person. For example, S2 had negative mu laterality (-0.203 dB) even

698 though both its contralateral (+1.491 dB) and ipsilateral (+1.694 dB) means were positive.
699 Separate session, hand and flagged/unflagged summaries are available in the verified
700 supplementary tables as descriptive sensitivity checks; they do not replace the all-trial population.
701 Table 8. Baseline-relative hand power and signed C3/C4 laterality.

Band	Contra. mean (dB)	Ipsi. mean (dB)	Mean L (dB)	Person L range (dB)	L signs (-/+)
Mu, 8–13 Hz	−0.308	−0.058	−0.250	−1.432 to +0.336	6 / 3
Beta, 13–30 Hz	−0.439	−0.272	−0.168	−0.519 to +0.133	8 / 1

702 Each of nine people receives equal weight, using equal-session mean trial dB from both hands.
703 Contra. and ipsi. average the corresponding C3/C4 hand components; L is contra. minus ipsi. All
704 2,592 hand trials, including 246 flagged trials, are retained. Negative L means a lower contralateral
705 than ipsilateral task/baseline change; it does not establish an absolute contralateral power decrease.
706 Values are rounded to three decimals here and retained at full precision in the data tables. These
707 fixed-estimator ratios are descriptive, not unbiased ERD estimates.
708 For the designated broad-EEGNet endpoint, descriptive associations between signed laterality and
709 saved binary LOSO balanced accuracy were $\rho = -0.150$ for mu and $+0.600$ for beta. Shared-
710 input coefficients were -0.083 and $+0.483$, and CSP4+LDA coefficients were $+0.067$ and $+0.317$,
711 respectively (Table S4; S1 Appendix, Fig S3). All six coefficients use the same nine people and
712 have no p-values. The positive beta coefficient indicates that higher balanced accuracy
713 accompanied a larger, less-negative signed descriptor; it does not support a benefit from stronger
714 relative contralateral suppression.

715 Discussion

716 Training-duration Sensitivity in a Small Development Benchmark

717 Training duration was a major source of internal performance variation. At two epochs, S3
718 remained near chance with either raw or source-normalized inputs; at 16 epochs, both performed
719 substantially better. Rank aggregation changed the source-selected schedule and recovered
720 performance for S3 and S8. The matched-runtime control also retained a large mean difference
721 between fixed training and the historical mean-loss schedule. An apparent cross-person model
722 collapse should therefore be examined alongside selected duration, source-validation trajectories,
723 per-class recall and prediction concentration. These outputs help distinguish an inadequate
724 stopping schedule from a failure attributed prematurely to the input representation.

725 These findings do not establish rank selection as a general solution. Within-fold ranks remove
726 loss-scale differences but discard the size of those differences; equal fold weighting imposes a
727 further aggregation choice. The candidate was developed after this benchmark had been inspected.
728 In addition, the matched-runtime fixed-duration condition improved the mean while the median
729 paired difference was negative. The external evaluations did not compare rank with mean-loss
730 selection and cannot independently confirm the internal gain. A prospective test would fix the
731 competing selection rules before observing new-person outcomes and match runtime, source
732 examples and update budgets.

733 Complete Spectral Pipelines Have Cohort-dependent Outcomes

734 The frozen evaluations limit claims of a consistent shared-input advantage. Shared μ/β input
735 gave no internal advantage, a small uncertain positive estimate on PhysioNet, a corrected adverse
736 contrast on Cho2017 and an uncertain near-zero contrast on Lee2019. The Cho2017 finding
737 contradicts a consistent advantage of this fixed shared-input pipeline under the evaluated source-
738 only adapter. It does not show that μ/β rhythms lack information, that parameter sharing is
739 generally harmful or that a different band definition would perform better.

740 The class recalls reveal a trade-off hidden by the cohort means. On Cho2017, the shared-input
741 pipeline reduced mean left-hand recall while increasing right-hand recall; on Lee2019, opposing
742 recall changes partly offset one another. These patterns describe the frozen predictions. They do
743 not justify correcting labels after inspection or attributing the contrast to physiology. Montage,
744 acquisition reference, amplitude scale, temporal alignment and band-logit averaging are possible
745 contributors, but the study did not isolate any of them.

746 One shared parameter set does not imply a cheaper computation: the network processes two band
747 views and pools their BatchNorm exposure. Independent-branch controls alter both capacity and
748 normalization, and the broadband capacity comparator retains a small disclosed parameter
749 mismatch. The source-fitted CSP model also transfers poorly and asymmetrically in Q15.
750 Performance above this contextual comparator does not establish practical or clinical reliability.
751 Contemporary adaptation and foundation-model methods were not compared under a common
752 budget, so the results support no state-of-the-art claim.

753 The external schedules are part of the pipeline contrast: broad/shared fits used 18/17 epochs in
754 Q14 and 14/19 in Q15. Paired differences therefore combine the selected training durations, input
755 views and shared normalization. They do not identify the causal contribution of parameter sharing
756 alone.

757 **Person-level Evidence and Descriptive Signal Characterization**

758 Additional seeds assess sensitivity to stochastic fitting; they do not increase the number of
759 participants. Lee2019 sessions and source-identity subsets likewise provide repeated observations
760 within a person. The paired person-level analysis retains all declared participants and reports the
761 direction of each difference. Its bootstrap intervals condition on the frozen source models and omit
762 uncertainty from another nine-person source cohort, acquisition adapter or development history.
763 For the internal study, overlapping outer training sets further limit an independent-person
764 interpretation.

765 A few large recoveries can improve a mean without producing consistent benefit across people, so
766 tests of average change and directional consistency need not agree. The original rank-selection
767 sign calculation retained exact ties; excluding them conventionally gives a different result from
768 both the archived calculation and the paired t-test. The discrepancy must remain visible rather than
769 be hidden under an incorrect test label. The later internal families retain their corrected null
770 findings. Percentile bootstrap intervals and randomization tests also need not agree in a small
771 sample; an interval excluding zero does not override a corrected null test. Q15 uses a separately
772 frozen two-cohort family: Cho2017 survives correction, whereas Lee2019 establishes neither
773 benefit nor equivalence. These separate tests do not establish a cohort-by-pipeline interaction.
774 PhysioNet has a different archived primary test and an amended protocol; its p-value belongs to
775 that evaluation rather than the newer family.

776 The BNCI signals describe measured sensor activity, not the cause of the decoder contrasts. Mean
777 signed hand laterality was negative in both bands: six of nine people had negative mu laterality
778 and eight had negative beta laterality, with positive signs in the others. A negative relative
779 descriptor can still accompany increased power on both sides, as in S2, so its absolute components
780 must also be examined. The positive broad-model beta correlation means higher balanced accuracy
781 accompanied less-negative relative laterality, rather than stronger suppression. None of the six
782 associations supports a subgroup threshold or a model-selection rule.

783 Neither a lateralized scalp mean nor its correlation with LOSO balanced accuracy locates a cortical
784 source, excludes ocular or muscle contributions, or demonstrates that EEGNet used that signal.
785 This BNCI-only physiology analysis was specified after decoder outcomes were known. It cannot
786 explain the adverse Cho contrast or validate an external physiological mechanism.

787 **Limitations**

788 Successive development conditions used the same nine BNCI people. Excluding the target from
789 each fit does not remove outcome-informed candidate design or dependence through overlapping
790 LOSO training sets. The internal rank-selection gain was concentrated in two people, and the

external evaluations did not test that selection-rule contrast. Controls also differ in runtime, training examples, update exposure, sensor width or approximate capacity. There is no common-budget comparison with contemporary adaptation or foundation-model methods.

Q14 and Q15 differ in montages, windows, references, filtering scope and source checkpoints. Q14's metadata amendment followed partial execution. In the frozen transfer experiment, repository freezes preceded fitting or inference but followed metadata inspection; they are not independently registered preregistration. Cross-provider participant identities remain unverified. The external bootstrap intervals condition on one frozen source cohort and omit source-sampling, adapter and development-history uncertainty.

The external duration schedules were selected using four fixed BNCI validation groups with unequal participant counts (2/2/2/3) and equal fold weighting. The grouping was frozen before target inference, but it is one arbitrary source-validation design. Its sensitivity was not tested prospectively.

Q15 used a common operational tensor and a committed adapter for transfer across three providers. Matching tensor dimensions does not make the physical measurements equivalent. Native export voltage calibration, the original Cho acquisition reference and hardware cue latency were not independently verified. The Cho channel order was mapped from the publication montage because the MATLAB payload has no channel-name strings. Lee event indexing retains a one-native-sample convention discrepancy. Numerical consistency checks cannot resolve these acquisition uncertainties.

Q14 and Q15 also differ in the extent of verification. The earlier PhysioNet evaluation required a sampling-rate amendment after partial execution; its later portability check examined saved outputs without replaying inference. Q15 froze preprocessing before source fitting and the inference contract before external prediction, and its completed scientific validator reports independent raw-to-prediction replay. The manuscript checks recalculate saved outputs and verify

816 hash bindings. These provide different levels of reproducibility and should not be treated as
817 interchangeable.

818 Evaluation was offline and used zero-phase filtering. Online latency, prospective assistive control,
819 clinical populations and patient benefit were not tested. The two-second Q15 crop omits part of
820 each imagery interval and differs from the providers' original benchmarks. A subsequent study
821 could preregister a calibrated adapter and frozen source-only models for another cohort, specifying
822 per-class and per-person outcomes before access. Retuning on the present held-out cohorts would
823 be further development, not independent external confirmation.

824 The physiology estimates compare a one-segment baseline with a three-segment task estimator;
825 unequal log-power variance can bias the dB difference. Negative signed laterality can arise from
826 ipsilateral power increases without absolute contralateral suppression. The descriptor and decoder
827 use different task windows, and nine descriptive person records with overlapping LOSO fits
828 cannot establish a subgroup biomarker or mechanistic attribution. Baseline-relative ratios cancel
829 only a constant multiplicative gain; they do not correct reference, timing, spatial mapping or
830 artifacts. External voltage calibration remains unknown, and external physiology has not been
831 evaluated.

832 **Conclusions**

833 Training-duration selection was consequential in this nine-person BNCI development benchmark,
834 but its gains were concentrated rather than broadly shared. Fixed shared-input pipelines showed no
835 consistent external advantage: balanced accuracy was 1.513 pp lower than broad EEGNet on
836 Cho2017, and the Lee2019 estimate remained compatible with small effects in either direction.

837 Unequal source-selected durations are part of these complete-pipeline comparisons, which do not
838 isolate parameter sharing. The post-outcome BNCI sensor analysis found negative mean signed
839 hand laterality in mu and beta with substantial participant variation; it established neither a

840 decoder mechanism nor external physiological replication. Unknown acquisition calibration and
841 offline evaluation limit deployment conclusions.

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922 **Supporting information captions**

923 S1 Appendix. Supplementary results, descriptive associations, inventories and detailed
924 reproducible methods. Internal supplementary figures and tables retain their original S numbering.

925 S1 Data. Frozen participant-level summaries, saved external predictions, descriptive physiological
926 outputs, source-data provenance and validation evidence. Files are CSV/JSON or gzip-compressed
927 CSV; raw EEG and model weights are excluded.